

HEMCHANDRACHARYA NORTH GUJARAT UNIVERSITY, PATAN

"Accredited By NAAC with 'A' Grade (CGPA 3.02)"

Bachelor of Computer Application (BCA)

w. e. f. June 2025 under NEP 2020

SEMESTER - V

COURSE TYPE	COURSE CODE	COURSE TITLE	CREDIT	WORK HOUR/WEEK		EXAM HOUR	TOTAL MARKS	
				TH	PR		TH	PR
Discipline Specific Course (MAJOR)	MS23MJDSCBCA501	JAVA Programming	4	4	-	2	100	-
Discipline Specific Course (MAJOR)	MS23MJDSCBCA501A	GUI Programming Using C# .Net	4	4	-	2	100	-
Discipline Specific Course (MAJOR)	MS23PMJDSCBCA501B	Practical: JAVA Programming	2	-	4	2	-	50
Discipline Specific Course (MAJOR)	MS23PMJDSCBCA501C	Practical: GUI Programming Using C# .Net	2	-	4	2	-	50
Minor Stream Course	MS23MIDSCBCA502	Software Engineering	4	4	-	2	100	-
Minor Stream Course	MS23MIDSCBCA502A	Operating System	4	4	-	2	100	-
Skill Enhancement Course	MS23SECBCA506	Project Development	2	-	2	2	-	50
Total			22	26			550	

Examination and Passing Criteria

- Internal Examination Marks Ratio: 50% of Total Marks.
- External Examination Marks Ratio: 50% of Total Marks.
- Passing Marks: 36% Marks in Internal as well as External Examination.

Evaluation System

The Evaluation System Consist of two components:

1. **Continuous and comprehensive Evaluation (CCE) Formative (Internal)**
2. **Semester End Evaluation (SEE) – Summative (External)**

In, each course 50% marks is assign to CEE and rest of 50 % marks is Assign to SEE.

Continuous and comprehensive Evaluation (CCE):

- The 50% marks assign to CEE is distributed between the continuous classroom evaluation and mid-term evaluation.
- Subject-wise CCE will be undertaken by the concerned faculty member. The mode of evaluation will be decided by the faculty member concerned with the subject.
- Normally CCE consists of **class participation, case analysis and presentation, assignment, tutorials, slip tests (announced/surprised), quizzes, attendance etc. or any combination of these.**
- The students are expected to submit their answer scripts/ reports of internal evaluation within the stipulated time. Failure to do so may result in the script not being valued.
- Another part of **CCE consists of mid-term written evaluation, which is compulsory for all students.** It can be done in a scheduled manner. The duration of the mid-term evaluation shall be one hour.
- Based on the types of evaluation, various models of evaluation implementation are suggested for theory, practical, self-study and work-based learning. The focus of these models is to encourage the students to improve on skills and performance.

Examination Pattern	Marks
Class Test (best 2 out of 3)	15
Quiz (Best 3 out of 4)	15
Active Learning	05
Class Assignment	05
Home Assignment	05
Attendance	05

Semester End Evaluation (SEE):

- The SEE carries 50% of the marks assigned to a course.
- SEE shall be of 2 hours for 4 credit course and 1 hour in case of 2 credit courses.
- The controller of the examination will conduct these examinations.
- Paper setting and evaluation will be done by the external examiners to an extent of 50% of the evaluation process.
- This examination shall be conducted as per a schedule which shall be notified in advance.
- The backlog exam will be conducted twice a year just after the result declared of the semester evaluation. Students shall have a second chance to clear their backlog and avoid the burden to carry forward the backlog with the next semester exam.

Eligibility Criteria to appear in SEE:

1. Should have at least 75% of attendance in all the courses put together
2. Should have at least 70% of attendance in each course/subject
3. Should not have any disciplinary proceedings pending against him/her
4. Should have no pending due

Exit Options

1. After II Semester - Exit option with Certificate in Computer Applications.
(With a minimum of 44 + 4 credits of Summer Internship)
2. After IV Semester - Exit option with Diploma in Computer Applications.
(With a minimum of 88 + 4 credits of Summer Internship)
3. After VI Semester - Exit Option with Bachelor of Computer Applications Degree, BCA Degree.
(With a minimum of 132 credits)
4. After VIII Semester - Award of Bachelor of Computer Applications Hons Degree, BCA (Hons.) Degree.
(With 176 credits)

Semester: V	Program Code:MGTUG201
Course Code: MS23MJDSCBCA501	Course Title: JAVA Programming
Course Credits: 04	Hours/Week: 04
Exam Duration: 2 Hours	Course Type: Discipline Specific Course (MAJOR).
Internal Exam Marks: 50	External Exam Marks: 50

Course Outcome:

After Completion of course,

- Students will learn to apply Object-Oriented Programming concepts effectively, using classes, inheritance, and polymorphism to develop robust Java applications.
- They will master the basics of Java programming, including understanding Java's structure, syntax, control statements, and exception handling, which will enable them to write and debug Java programs.
- Additionally, students will gain skills in advanced Java topics, such as implementing interfaces, managing packages, working with multithreading, and creating applets and graphical user interfaces.

Total Teaching Hour: 45		
Sr. No.	PARTICULAR	MARKS
Unit - I	Fundamentals of Object Oriented Programming: Basic Concepts of Object-Oriented Programming: Object and Class, Data Abstraction and Encapsulation, Inheritance, Polymorphism, Dynamic Binding, Message Communication. Benefits and Applications of OOP. Basics of Java. History of Java, Java Features, JDK and its Components (Various Tools of JDK), Byte code and JVM, JAVA Program Structure. Building Blocks: Tokens (Keywords, Identifier, Literals, Operators, Separators), Variables, Data Types, Type Conversion and Casting, Command Line Arguments. Control Statements: Selection Statement: if – else, nested if-else, else if ladder, switch Statements. Iteration statements: while, do...while, for Statements Jump Statements: break, continue. Arrays: How to create array in java.	25%
Unit - II	Class Fundamentals: Defining Classes, Creating Objects, Static Members, Methods Overloading, Nesting of Methods Constructors: Introduction, Default & Parameterized Constructors, Constructor Overloading. Inheritance: Introduction, Single, Multilevel, Hierarchical Inheritance. Method Overriding, final Variable, final Methods & final Class, abstract Methods and Abstract Class. Access Modifiers (Visibility Controls) – public, private, default, protected, private protected.	25%

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Unit - III	Interfaces: Defining an Interfaces, Implementing Interfaces, Multiple interfaces. Packages: Defining Package, List of Java API Packages and their classes, Creating, Accessing and Using packages, adding a Class to Package. Java.lang Package classes (Math, String). Java.util Package classes (Stack, Vector). Multithreading : Definition, Life Cycle of Thread, Extending Thread Class, Implementing Runnable Interface, Various Thread Methods like yield(),stop(),sleep(), Thread Priorities.	25%
Unit - IV	Exception Handling : Exception & Exception Handling, Exceptions Types (Java's Built-in Exceptions) – Checked and Unchecked Exceptions, Exception Handling using try, catch, finally, throw, throws etc., Nested try, Multiple catch, Throwing Your own Exception. Applets: Introduction, Local and Remote Applets, Applet Lifecycle, Creating and executing Applet, Passing Parameters to Applet. Graphics Class: Various methods of Graphics Class.	25%

Text & Reference Books:

1. Programming with JAVA, E Balagurusamy, Tata McGraw Hill
2. The Complete Reference JAVA, Herbert Schildt, Tata McGraw Hill .
3. Teach Yourself JAVA. Joseph O'Neil & Herb Schildt, Tata McGraw Hill .

University Question Paper Scheme			
Q.1	Unit-I	Descriptive/ Long questions with choice	10 Marks
Q.2	Unit-II	Descriptive/ Long questions with choice	10 Marks
Q.3	Unit-III	Descriptive/ Long questions with choice	10 Marks
Q.4	Unit-IV	Descriptive/ Long questions with choice	10 Marks
Q.5	All Unit	Objective / Short Question / True –False etc.	10 Marks

Semester: V	Program Code: MGTUG201
Course Code: MS23MJDSBCA501A	Course Title: GUI Programming Using C# .net
Course Credits: 04	Hours/Week: 04
Exam Duration: 2 Hours	Course Type: Discipline Specific Course (MAJOR).
Internal Exam Marks: 50	External Exam Marks: 50

Course Outcome:

After Completion of course, Student able to

- Gain proficiency in using **Visual Studio tools** for creating and managing Windows applications.
- Master the principles of **Object-Oriented Programming (OOP)** using C#.
- Develop practical skills in working with **.NET controls**.
- Understand and apply **ADO.NET** for database manipulation with databases.
- Learn the fundamentals of **Crystal Reports** to design and deploy professional reports.

Total Teaching Hour: 45		
Sr. No.	PARTICULAR	MARKS
Unit - I	Basic of the .Net Framework : What .net technology?, Comparison of .net technology over other current technology, Advantages of .net technology, Comparison & Overview of all Frameworks, Introduction about applications to be developed by .net. .Net Framework Architecture : The .Net Framework Architecture, Role of CLR in .net framework, Introduction about language supported by .net, CLR architecture, Managed and unmanaged code, Compiling Source code into Managed module, Introduction about class library (FCL), MSIL code. Role of assembly and meta data., JIT compiler and its types, JIT vs Traditional compiler, Class Loader, Name space (Purpose and Types), CTS – Value types and reference types, CLS, Interoperability with unmanaged code.	25%
Unit - II	Introduction about Visual studio Tool Comparisons and overview with all visual studio Tools versions. Create new windows application with vb.net, C sharp and other language. Introduction about all categories of toolbox control. Design window, solution explorer, Server Explorer, Property and Event Explorer, Class view, Command window, Introduction about code window, Understand how to write code, run application, debugging application. Introduction about menus and functionalities of all menu bar available in Tools. Class and Event driven model. Basics of C# Advantages of OOPS. Creating Class and Object, Structure of Class. Data types, Operators, Constructor, Destructor, Abstraction, Interface, Polymorphism (Overloading, overriding), Inheritance, Garbage collection, Jagged Array, Collection (hasht table and Array List), Indexer (One dimension) and property, Delegates and Event Multicasting), Exception handling, String handling functions, Creating function with all types.	25%

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Unit - III	Understanding Controls Net Common Controls: Control Hierarchy, Label and Link Label, Text Box and Rich Text Box, Picture Box, Button, Group Box, Panel, Check Box and Radio Button, List Box, Checked List Box and Combo Box, Month Calendar and Date Time Picker, Tree View and List View, Timer, Track Bar and Progress Bar, Image List control, HScroll Bar, VScroll Bar, Tab Control. Common Dialogs Control: Color Dialog, Folder Browser Dialog, Font Dialog, Open File Dialog, Save File Dialog, MDI-Forms, Exploring Properties, Methods and Events, Menu bar, Context Menu, Message box, Input box.	25%
Unit - IV	ADO.Net Architecture Of ADO.Net, Data Base Manipulation, .Net Data Provider, Data Adapter, Data Set, Data Table, Introduction about SQL server. Connection with SQL server. Command, Data Reader, Data Grid View, Execute reader, Execute Non Query, Execute Scalar. Crystal Reports Syllabus: Introduction to Crystal Reports and its integration with Visual C#, Designing reports and working with data sources, Formatting, grouping, Adding parameters, and summarizing report data, Exporting and deploying reports in different formats. Setup and Deployment of .net Applications.	25%

Text & Reference Books:

1. The complete reference c sharp 4.0 By, Herbert Schildt, Publisher: McGraw-Hill
2. Microsoft Visual C# 2010 Step by Step, By John Sharp
3. Microsoft. Professional C# 5.0 And .Net 4.5.1, By Christian Nagel, Jay Glynn, Morgan Skinner
Publisher: Wrox

University Question Paper Scheme			
Q.1	Unit-I	Descriptive/ Long questions with choice	10 Marks
Q.2	Unit-II	Descriptive/ Long questions with choice	10 Marks
Q.3	Unit-III	Descriptive/ Long questions with choice	10 Marks
Q.4	Unit-IV	Descriptive/ Long questions with choice	10 Marks
Q.5	All Unit	Objective / Short Question / True –False etc.	10 Marks

Semester: V		Program Code: MGTUG201
Course Code: MS23MJDSBCA501B	Course Title: Practical – JAVA Programming	
Course Credits: 02	Hours/Week: 04	
Exam Duration: 2 Hours	Course Type: Discipline Specific Course (MAJOR).	
Internal Exam Marks: 25	External Exam Marks: 25	

Total Teaching Hour: 45**Practical List**

1. Write a Java Program find the area of circle.
2. Write a Java Program that will display factorial of the given number.
3. Write a Java Program that will find the largest no from the given two nos.
4. Write a Java Program that will find the largest no from the given three nos.
5. Write a Java Program that shows the use of switch Statement.
6. Write a Java Program to find the sum of the digits of given number.
7. Write a Java Program that will display the Sum of $1+1/2+1/3+...+1/n$.
8. Write a Java Program that checks whether the given no is prime or not.
9. Write a Java Program that implements the use of break statement.
10. Write a Java Program that implements the use of continue statement.
11. Write a Java Program that will accept Command-line Arguments and display the same.
12. Write a Java Program to sort the elements of an array in Ascending Order.
13. Write a Java Program to create a Student class and generate result of student (Total, Per, Grade).
14. Write a Java Program to create an Employee class and generate Salary Slip for the employee.
15. Write a java program which shows the use of Static Members.
16. Write a java program which shows the Nesting of Methods.
17. Write a java program which shows the use of Methods Overloading.
18. Write a java program which implements the Default Constructors.
19. Write a java program which implements the Parameterized Constructors.
20. Write a java program which implements the Overloading of Constructors.
21. Write a java program which explains the concept of Single Inheritance.
22. Write a java program which explains the concept of Multilevel Inheritance.
23. Write a java program which explains the concept of Hierarchical Inheritance.
24. Write a java program which shows the Method Overriding.
25. Write a Java Program to implement final class and final method.
26. Write a Java Program to implement abstract class and abstract method.
27. Write a java program which implements Interface.
28. Write a java program which implements Multiple Interfaces.
29. Write a java program which shows importing of classes from other packages.

30. Write a Java Program to implement the methods of Math Class.
31. Write a Java Program to implement the methods of String Class.
32. Write a Java Program to implement the methods of Vector Class.
33. Write a Java Program to implement the methods of Stack Class.
34. Write a Java Program which will read a text and count all occurrences of a particular word.
35. Write a Java Program which will read a string and rewrite it in the alphabetical order.
eg.The word "STRING" should be written a "GINRST".
36. Write a java program which creates threads using the Thread Class.
37. Write a java program which shows the use of yield(), stop() and sleep() Methods.
38. Write a java program which shows the Priority in Threads.
39. Write a java program which use of Runnable Interface.
40. Write a java program which uses try and catch for Exception Handling.
41. Write a java program which uses Multiple catch Blocks.
42. Write a java program which uses finally Statement.
43. Write a java program which uses Nested try Statements.
44. Write a java program which shows throwing our own Exception.
45. Create an Applet program that print Hello Applet.
46. Create an applet that use init(),start(),stop() and destroy() methods of applet.
47. Write an applet program to implement the concept of passing parameter to applet.
48. Write a applet program to implement various methods of Graphics class.

Note: Note: In the external examination, programs will be provided by the external examiner and may not be selected from the practical list.

University Practical Examination Scheme:

Examination Duration: **2 Hours (Per Batch)**

Practical Marks: **10** Viva Marks: **10** Journal Marks: **5**

Semester: V		Program Code: MGTUG201
Course Code: MS23MJDSCBCA501C	Course Title: Practical – GUI Programming	
Course Credits: 02	Hours/Week: 04	
Exam Duration: 2 Hours	Course Type: Discipline Specific Course (MAJOR).	
Internal Exam Marks: 25	External Exam Marks: 25	

Total Teaching Hour: 45**Practical List**

1. Develop a console application that prompts the user to enter various values like Name, Age, and City. Once the user inputs the data, display it back on the console.
2. Write a program that takes a number between 0 and 999 from the user and converts it into words. For example, if the user enters 98, the program should output "Ninety-eight". Consider special cases like zero and numbers with exact hundreds.
3. Create a program that asks the user for a main string and a substring. The program should then count how many times the substring occurs within the main string and display the result.
4. Define a class called Person with properties like Name and Age, and a method ShowDetails(). The program should create objects of the Person class, assign values to the properties, and call the method to display the details of each person.
5. Write a console program that demonstrates constructor overloading by creating a class Rectangle with multiple constructors (one for width and height, one with default values). Also, implement method overloading by creating multiple CalculateArea() methods that accept different parameters.
6. Create a class Animal with a virtual method MakeSound(). Derive two classes Dog and Cat from it, overriding the MakeSound() method in each. Instantiate the derived classes and demonstrate method overriding by calling MakeSound() on each object.
7. Write a program that demonstrates various string manipulation functions such as Substring, Replace, ToUpper, ToLower, and Split. Ask the user for a string input, then apply each function and display the results.
8. Create a console application that uses exception handling to prevent runtime errors. The program should include a try, catch, and finally block to handle cases like dividing by zero or accessing invalid array indices.
9. Write a program that defines a delegate Calculate which can point to methods for arithmetic operations like addition and multiplication. Demonstrate multicasting by allowing the delegate to call both methods sequentially and display the results.
10. Develop a console application that uses a jagged array to store and display values. For example, it can store different numbers of grades for different students, then print each student's grades using nested loops.
11. Create a Windows form that includes labels and textboxes for First Name, Last Name, and

- Email. Add a Register button that, when clicked, displays the entered data in a message box.
12. Develop an application with a textbox and three buttons labeled Left, Right, and Center. Clicking each button should adjust the text alignment within the textbox according to the selected button.
 13. Create a Windows form that includes two TextBox controls for Username and Password, along with labels indicating where the credentials should be entered. Include a Login button that validates the inputs and displays an error message if either the username or password is empty.
 14. Design a form that contains a GroupBox with several buttons. Each button will perform a different action, such as changing the form's background color, displaying a message box, or clearing all input fields on the form.
 15. Create a Windows form with a ComboBox listing various countries. When a country is selected, the name of the selected country should be displayed on a label.
 16. Develop a simple calculator application that performs basic arithmetic operations like addition, subtraction, multiplication, and division. Use buttons for numbers and operations, and a TextBox to display the result. Design the form to clearly display buttons in a grid layout for ease of use.
 17. Create a form where users can select hobbies using CheckBox controls. When the user clicks a Submit button, all selected hobbies should be added to a ListBox.
 18. Write a program that allows users to enter a series of numbers separated by commas into a textbox. Create a function that retrieves these numbers, checks whether each is odd or even, and stores the odd numbers in one ListBox and the even numbers in another. When the user clicks a button, the program should process the input and display the results in the respective ListBoxes.
 19. Develop an application where users can input a sentence into a TextBox. The program will analyze the sentence and display the number of vowels, spaces, digits, special symbols, and words. Additionally, check if the sentence is a palindrome and display the result.
 20. Create a To-Do List application where users can add tasks to a ListBox using a TextBox and an Add button. Users should be able to select and remove tasks from the ListBox using a Remove button.
 21. Create a form that contains four RadioButton controls for selecting gender (Male, Female) and marital status (Married, Unmarried). When a button is clicked, display the selected gender and marital status in a message box.
 22. Design a Windows form with a DateTimePicker control and a label. When the user selects a date from the DateTimePicker, display the selected date in the label.
 23. Create a form with two MonthCalendar controls and a Calculate button. When the user selects two dates and clicks the button, the program should display the difference between the two dates in days. This demonstrates date selection and simple date calculations.

24. Develop a form where users can choose a shape (like a circle or rectangle) using radio buttons. Once the user selects a shape, the chosen shape should be displayed on the form using simple drawing logic.
25. Create a form with a button that opens a ColorDialog. When the user selects a color, apply that color as the background color of the form.
26. Write a program that uses an ImageList to store multiple images. The form will include a TabControl with multiple tabs, each tab displaying a different image from the ImageList. The user can switch between tabs to view the images.
27. Create an application that includes a TrackBar control linked to a ProgressBar. As the user slides the TrackBar, the ProgressBar should reflect the progress corresponding to the current value of the TrackBar.
28. Design a stopwatch application using a Timer control. The program should include Start, Stop, and Reset buttons to control the timer, and it should display the elapsed time in a label or TextBox.
29. Develop a form that uses the SplitContainer control to separate the form into two sections. Place a TreeView control in one section to represent a directory structure, and a ListView control in the other section to display files from the selected directory. When a node in the TreeView is clicked, the files from that directory are displayed in the ListView.
30. Design an application that includes a ListBox and a ContextMenuStrip. When the user right-clicks on an item in the ListBox, the ContextMenuStrip should display options like Edit and Delete, and perform the selected action.
31. Create an MDI form that includes a File menu with options for Open, Close, and Exit. Users can open an image file using an open dialog box, and the image will be displayed in a PictureBox. The child form will include buttons that allow the user to zoom in, zoom out, and reset the image to its original size.
32. Write a program that acts as a simple text editor. The user can use an OpenFileDialog to open a text file, edit it in a RichTextBox, and save the edited text back to the file system using a SaveFileDialog. The program should also allow basic text editing functions like Copy, Paste, and Undo.
33. Develop a database application using ADO.NET with SQL Server that allows users to perform Select, Insert, Update and Delete operations on a table named EMP. The table contains fields like EMP_ID, EMP_NAME, DOJ, GENDER, ADDRESS, MARITAL STATUS and Mobile. The form should include proper validation and use a DataGridView for display records. Perform navigation through buttons.
34. Develop a database application using ADO.NET with MS Access that allows users to perform Select, Insert, Update and Delete operations on a table named REGISTRATION. The table contains fields like ID, NAME, DOB, QUALIFICATION, GENDER, ADDRESS, MARITAL STATUS and Mobile. The form should include proper validation and use a DataGridView for display

records. Perform navigation through buttons.

35. Design a form with a TextBox for inputting search criteria. When the user types a value in the TextBox, the program should query the database and retrieve matching records. The results should be displayed in a DataGridView. The search should support partial matches and update the DataGridView dynamically based on the search input.
36. Write a program that accepts a User ID and Password, and validates them against values stored in a database. The program should display whether the credentials are valid or not, based on the database check.
37. Design a form to manage records for a Salesman table with fields such as ID, Name, Gender, Sales Date, Items, Quantity, and Total Amount. The form should allow users to add sample records and generate a crystal report that displays records based on the selected sales date.
38. Design a form to manage records for a Salesman table with fields such as ID, Name, Gender, Sales Date, Items, Quantity, and Total Amount. The form will also include two DateTimePicker controls for selecting a start and end date, and RadioButton controls for selecting gender. Users should generate crystal report based on the selected date range and gender.

Note: In the external examination, programs will be provided by the external examiner and may not be selected from the practical list.

University Practical Examination Scheme:

Examination Duration: **2 Hours (Per Batch)**

Practical Marks: **10** Viva Marks: **10** Journal Marks: **5**

Semester: V		Program Code:MGTUG201
Course Code: MS23MIDSCBCA502	Course Title: Software Engineering	
Course Credits: 04	Hours/Week: 04	
Exam Duration: 2 Hours	Course Type: Minor Stream Course	
Internal Exam Marks: 50	External Exam Marks: 50	

Course Outcome:

After Completion of course,

- Students will understand core software engineering concepts, including software characteristics, process models (Waterfall, Prototyping, Incremental, Spiral), and software quality factors.
- They will gain skills in requirements specification, system design strategies, coding practices, and testing methods.
- Additionally, students will learn about software project management, including scheduling, quality assurance, and risk management, along with software reliability and metrics.

Total Teaching Hour: 45		
Sr. No.	PARTICULAR	MARKS
Unit - I	Introduction to Software Engineering: Define Software and System; Define Software Engineering, Software Characteristic, Difference between Software Engineering and Computer Science. Difference between Software Engineering and System Engineering. Software Costs, Software Application, Evolution of software Engineering, Software Crisis-Problem and Causes, Software Myths, Professional and Ethical Responsibility, Software Process, Principal of Software Engineering, Software Quality Factors, Software Quality Attributes, Software Engineering Methods. Software Process Model : Waterfall Model, Prototyping Model, Incremental Model, Spiral Model	25%
Unit - II	Software Requirement Specification What is Requirement, Types of Requirement, SRS (Software Requirement Specification), Software Engineering Benefits, Role of Management in Software Development, Role of Metrics and Measurement. System Design Software Design Strategy, Become a Master Designer, Evaluating a Design, Problem Partitioning, Abstraction, Strategy of Design, Function Oriented v/s Object Oriented Approaches	25%
Unit - III	Coding Programming Practices, Top down Approaches & Bottom up Approaches, Structure Programming, Information hiding, Programming Style Testing Testing Fundamental, Top Down Approaches & Bottom Up Approaches, Test Cases and Test Criteria, Psychology of Testing, Regressing Testing, Functional Testing, Structure Testing. Equivalence Class Partitioning, Boundary Value Analysis, Cause Effect Graphing, Type of Testing, Test Plan.	25%

Unit - IV	Software Project Management Software Project Management, COCOMO Model, Project Scheduling, Software Configuration Management, Software Maintenance, Quality Assurance Plans, Verification and Validation, Project Monitoring Plans, Software Risk Management, Project Planning. Software Reliability and Quality Assurance Software Reliability, Reliability metrics, Objectives of Software quality Management, ISO-9000 Certification for Software industry, Software measurement and metrics, SEI Capability Maturing Model, Computer Aided Software Engineering SASE and its Scope, Software Reverse Engineering, What is Risk, Risk Management.	25%
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Text Book:

1. **Software Engineering: A Practitioner's Approach**, Roger S. Pressman, 8th Edition, McGraw-Hill Education.

Reference Books:

1. Software Engineering, Ian Sommerville, 10th Edition, Pearson.
2. Introduction to Software Engineering Design, Christopher Fox, 1st Edition, Addison-Wesley.
3. Object-Oriented Software Engineering Using UML, Patterns, and Java, Bernd Bruegge, Allen H. Dutoit, 3rd Edition, Pearson.
4. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI Learning.
5. Applied Software Project Management, Andrew Stellman, Jennifer Greene, 1st Edition, O'Reilly Media.

University Question Paper Scheme			
Q.1	Unit-I	Descriptive/ Long questions with choice	10 Marks
Q.2	Unit-II	Descriptive/ Long questions with choice	10 Marks
Q.3	Unit-III	Descriptive/ Long questions with choice	10 Marks
Q.4	Unit-IV	Descriptive/ Long questions with choice	10 Marks
Q.5	All Unit	Objective / Short Question / True –False etc.	10 Marks

Semester: V		Program Code:MGTUG201
Course Code: MS23MIDSCBCA502A	Course Title: Operating System	
Course Credits: 04	Hours/Week: 04	
Exam Duration: 2 Hours	Course Type: Minor Stream Course	
Internal Exam Marks: 50	External Exam Marks: 50	

Course Outcome:

After Completion of course, Students will

- Understand the foundational concepts of operating systems, including their roles, types, and core services, and apply this knowledge to differentiate between various operating system architectures.
- Analyze and manage processes through different scheduling algorithms and handle deadlocks, enhancing problem-solving skills in process management.
- Implement and manage memory and file systems using virtual memory concepts, file management techniques, and UNIX commands to efficiently administer and troubleshoot operating systems.

Total Teaching Hour: 45		
Sr. No.	PARTICULAR	MARKS
Unit - I	Introduction Overview of Operating Systems: Definition, role and Goal of an Operating System, Components of Operating System, Services of Operating System Types of OS: Batch OS, Multiprogramming OS, Time-sharing OS, Multiprocessing OS, Distributed OS, Real-time OS, Mobile OS and Embedded OS. Additional Concept: Buffering, Spooling, caching	25%
Unit - II	Process Management Process Concepts: Definition of a process, Process control block (PCB), Process states Transition Diagram, Context switching. Schedulers: Long-term schedulers, Medium-term schedulers, Short-term schedulers. Process Scheduling Algorithms: FCFS, SJF, Priority Scheduling and Round-Robin (All With Examples), Overview of Multilevel Queue and Multilevel Feedback Queue Scheduling. Threads: Definition, Benefits, Difference between threads and processes. Deadlock: Definition, characteristics. Scheduling criteria: Throughput, Turnaround time, Response time, Waiting Time, CPU burst.	25%

Unit - III	Memory Management Basic Concepts: Contiguous memory allocation, Logical vs. physical address space. Partitioning: Fixed Partitioning, Dynamic Partitioning. Fragmentation and Defragmentation (Compaction) Virtual Memory: Concept and benefits of virtual memory, Paging, Demand paging. Page replacement: Introduction, Steps for Page Replacement, Page replacement Algorithms - FIFO, Optimal, LRU I/O Management: Program controlled I/O and Interrupt Driven I/O	25%
Unit - IV	UNIX Introduction UNIX Overview: Features of Unix, Kernel. Unix File Management: File Types, File Permissions, Access Methods, Directory Structure. Basic Commands: Man, pwd, cd, ls, mkdir, rmdir, cp, cat, more, mv, rm, chmod, diff, grep, wc, Wildcards. Process Management Commands: Ps, Kill, Nice, Sleep, Batch, At	25%

Text Book:

1. **"Operating System Concepts"** by Abraham Silberschatz, Henry Korth, S. Sudarshan *Publisher:* John Wiley & Sons, 10th Edition.

Reference Books:

1. **"Modern Operating Systems"** by Andrew S. Tanenbaum *Publisher:* Pearson
2. **"Operating Systems: Internals and Design Principles"** by William Stallings *Publisher:* Pearson
3. **"UNIX Operating System"** by Sumitabha Das, 4th Edition, McGraw-Hill Education.
4. **"Understanding the Linux Kernel"**, Daniel P. Bovet, Marco Cesati, 3rd Edition, O'Reilly Media.
5. **"UNIX and Linux System Administration Handbook"** by Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley, Dan Mackin *Publisher:* Prentice Hall

University Question Paper Scheme			
Q.1	Unit-I	Descriptive/ Long questions with choice	10 Marks
Q.2	Unit-II	Descriptive/ Long questions with choice	10 Marks
Q.3	Unit-III	Descriptive/ Long questions with choice	10 Marks
Q.4	Unit-IV	Descriptive/ Long questions with choice	10 Marks
Q.5	All Unit	Objective / Short Question / True –False etc.	10 Marks

Semester: V		Program Code:MGTUG201
Course Code: MS23SECBCA506	Course Title: Project Development	
Course Credits: 02	Hours/Week: 02	
Exam Duration: 20 min. / Group	Course Type: Skill Enhancement Course	
Internal Exam Marks: 25	External Exam Marks: 25	

Course Outcome:

After Completion of course,

- Students will gain practical experience in project development, enhancing their problem-solving and coding skills.
- Students will learn to work collaboratively, managing all phases of a project from planning to implementation.
- Students will demonstrate their ability to develop functional, innovative projects and present comprehensive project documentation.

Total Teaching Hour: 25
PARTICULAR
<u>General Guidelines</u> ▪ The In Semester Project is designed to provide practical, hands-on experience in project development. ▪ The course will run throughout the semester, where students work in groups under the guidance of a project guide. ▪ The project work will culminate in an internal and external evaluation based on the project's functionality, innovation, and documentation. ▪ This subject is mandatory and plays a significant role in preparing students for real-world challenges. ▪ Students will form groups of 2 members (3 in specific cases with approval). Each group will be assigned a Project Guide at the start of the semester. ▪ Students will select a project topic that aligns with course goals and get approval from their project guide or the subject teacher. ▪ Internal Examination Conducted by the assigned project guide, and results will be submitted to the subject teacher. ▪ External Examination Conducted at the end of the semester, the external examiner will assess the projects, supported by documentation and functional demonstration. ▪ Subject teacher must maintain clear and comprehensive records of project development and progress. ▪ Students group has to develop a comprehensive project report to be submitted at the end of the semester. The report should cover all phases of development, including planning, implementation, testing, and results.

Guidelines for Subject Teacher

- Select one project at the start of the semester and develop it in front of the students, demonstrating the entire project development process from start to finish in entire semester.
- Break down the project into manageable modules, covering each stage during lectures, so students can understand the workflow, coding practices, and troubleshooting involved.
- Maintain records of each student group's project definition, attendance, project progress, and internal examination results.
- Offer regular guidance to students, addressing challenges they may face during development.
- Collection of internal examination results from project guides.

University Practical Examination Scheme:

Examination Duration: 20 minutes / Group

Project Work: **10** Viva Marks: **10** Documentation: **5**